

Rudra Pratap Singh

Full Stack Developer | AI Engineer

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Education

Indian Institute of Technology Roorkee, BS-MS in Mathematics and Computing

Nov 2021 - May 2026

- **Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Database Management System, Statistical Machine Learning, Analytics for Managerial Decision Making.

Experience

Lead Engineer (Full Stack)

[Milk. Inc](#)

June 2025 - Present

- Engineered an automated pipeline using `face_crop_plus` and advanced AI models (e.g., `birefnet-portrait`) to preprocess 500+ award photos with >98% accuracy, ensuring uniform alignment and background consistency.
- Built an app with customizable crop, background, and format options to automate image processing.
- Automated slide and panel generation via Python backend, replacing VBA for better maintainability and portability.

AI Engineer Intern

[DeeCogs Technologies Pvt. Ltd.](#)

February 2025 - Present

- Developed [PhysioTrack](#), a Python library for real-time human pose estimation using **MMPose** and **RTMLib**, analyzing joint angles to assess spinal mobility (flexion/extension) for physiotherapy applications.
- Engineered algorithms to calculate critical biomechanical angles from video/webcam feeds, enabling data-driven pain-point evaluation for patients.
- Built an AI physiotherapy agent using Google ADK for real-time video/audio analysis and automated feedback.

Projects

TruckNet

[Source Code](#)

| [Deployed Link](#)

March 2025 - April 2025

- Built a load bidding platform where shippers post loads, truckers bid live, and the lowest bid wins in real time.
- Built WebSocket-powered bidding rooms for real-time participation, live updates, and competitive load allocation.
- Developed secure RESTful APIs with JWT authentication and a SuperAdmin dashboard for approvals and oversight.

FoodieRAG

[Source Code](#)

April 2025

- Developed a RAG chatbot using LangChain, Hugging Face models, and Streamlit for restaurant-based Q&A.
- Automated restaurant data scraping and processing via Selenium, BeautifulSoup, and modular Python scripts.
- Built a FAISS vector database with sentence-transformer embeddings for efficient retrieval.

LangChain Email Generator

[Source Code](#)

| [Deployed Link](#)

January 2025

- Engineered a dynamic full-stack application to generate professional emails using Next.js and Tailwind CSS.
- Integrated LangChain and gemini-2.0-flash model to enable AI-driven email content generation.
- Implemented secure data storage with MongoDB and Mongoose while managing user sessions with cookies.

Technical Skills

- **Languages:** Python (Pandas, NumPy), C++, JavaScript/TypeScript, SQL
- **Frameworks:** React, Next.js, Node.js, Express, Django, PyTorch, TensorFlow
- **Tools:** Git, Docker, AWS, MongoDB, PostgreSQL, Linux

Achievements

- **All India Rank 4563, JEE Advanced 2021:** Achieved AIR 4563 in JEE Advanced 2021 out of 1,51,000 candidates.
- **Specialist on Codeforces:** Maximum Rating of 1551 on Codeforces (Handle - chipi-chipi_chapa-chapa).
- **National Finalist, Code With Cisco:** Ranked in the top 0.25% of 30,000+ participants nationwide.

Positions of Responsibility

Cup Captain

Nov 2023 - Jan 2024

- Led a contingent of 50+ members across Parliamentary Debates, Standup Comedy and Improv Comedy.
- Contributed to securing overall Second Runner Up in Speaking Arts across all IIT's in Inter-IIT Cult Meet 2023.